Portfolio Summary

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- Mary Ann Mastri
- Artificial Intelligence, Robotics and Cyber Security
- After spending 25 years in the trading world—primarily focused on crude oil trading and risk control—I reached a point where I needed a reset. I've returned to the classroom with genuine enthusiasm, diving into new technologies and expanding my understanding of the field. This journey has been both energizing and eye-opening, revealing how much there was to learn while I was fully immersed in my career. I'm now committed to deepening my expertise and building a strong foundation for the next chapter of my professional life.
- https://github.com/m-girl/NLP

My Learning Journey

- **Foundations:** Python, Data Analysis, ML Basics
- **Core ML:** Classification, Regression, Model Evaluation
- **Deep Learning:** Neural Networks, Embeddings
- **NLP:** Tokenization, Transformers, Topic Modeling
- **AI Systems:** Pipelines, Evaluation Suites, UI Integration
- **Final Project:** NewsBot 2.0
- **Technologies :**
  - Scikit-Learn
  - SpaCy
  - NLTK
  - Sentence-Transformers
  - Gradio
  - Pandas / NumPy

Featured Projects

- **Machine Learning**
  - Final project integrating all core ML concepts learned throughout the course
  - Demonstrated full pipeline development, evaluation, and deployment
- **Data Analytics**
  - Developed a Retrieval-Augmented Generation (RAG) tool called **Study Buddy**
  - Designed to support efficient learning and personalized study workflows
- **Natural Language Processing**
  - Built **NewsBot 2.0**, an end-to-end AI news intelligence system
  - Includes classification, topic modeling, sentiment analysis, semantic search, and conversational Q&A

Skills and Competencies

Technical Skills

- Machine Learning
- Natural Language Processing
- Semantic Search
- Model Evaluation
- Data Engineering
- Python Development

Core AI Competencies

- Classification & Topic Modeling
- Sentiment Analysis
- Named-Entity Recognition
- Summarization
- Embedding-based Retrieval
- System Integration

Soft Skills

- Problem-solving
- Communication
- Project Ownership
- Research & Experimentation
- Collaboration

Reflection & Next Steps

- **Reflection**
  - Built my first full AI system from scratch
  - Learned how to integrate multiple ML components
  - Gained confidence in NLP and semantic search
  - Improved debugging, testing, and evaluation skills
- **Next Steps**
  - Explore LLM-powered enhancements
  - Add multilingual summarization
  - Deploy NewsBot 2.0 as a web app
  - Continue building portfolio-ready AI system

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